

CLOUD COMPUTING

UE23CS351B

LAB 5- Kubernetes

*Scenario based – Robots
City*

GITHUB:

**[https://github.com/akshayag2005/CLOUD-
COMPUTING](https://github.com/akshayag2005/CLOUD-COMPUTING)**

DATE: 10/02/2026

NAME: AKSHAY

SRN: PES2UG23CS044

SECTION: A

SS1:

```
(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ kubectl create namespace pes2ug23cs044-alpha
namespace/pes2ug23cs044-alpha created

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ kubectl create namespace pes2ug23cs044-beta
namespace/pes2ug23cs044-beta created

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ kubectl get namespaces
NAME                                STATUS    AGE
default                            Active   2m
kube-node-lease                    Active   2m
kube-public                        Active   2m
kube-system                        Active   2m
pes2ug23cs044-alpha                Active   28s
pes2ug23cs044-beta                Active   8s

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$
```

SS2:

```
(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ kubectl run citizen --image=nginx --restart=Never
pod/citizen created

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ kubectl get pods
NAME    READY   STATUS    RESTARTS   AGE
citizen 0/1     ErrImagePull 0           18s

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ kubectl describe pod citizen
Name:         citizen
Namespace:    pes2ug23cs044-alpha
Priority:      0
Service Account: default
Node:         minikube/192.168.49.2
Start Time:   Tue, 10 Feb 2026 14:57:41 +0530
Labels:       run=citizen
Annotations:   <none>
Status:       Pending
IP:           10.244.0.3
IPs:          IP: 10.244.0.3
Containers:
  citizen:
    Container ID:
    Image:        nginx
    Image ID:
    Port:         <none>
    Host Port:    <none>
    State:        Waiting
      Reason:     ImagePullBackOff
    Ready:        False
    Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-jtzq4 (ro)
Conditions:
  Type           Status
  Initialized     True
  Ready           False
  ContainersReady False
  PodScheduled    True
Volumes:
  kube-api-access-jtzq4:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    Optional:            false
    DownwardAPI:         true
    QoS Class:           BestEffort
    Node-Selectors:      <none>
    Tolerations:         node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                        node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason      Age    From          Message
  ----    -
  Normal  Scheduled   26s    default-scheduler  Successfully assigned pes2ug23cs044-alpha/citizen to minikube
```

SS3:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG$ kubectl delete pod citizen
pod "citizen" deleted from pes2ug23cs044-alpha namespace

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG$ kubectl get pods -w
```

NAME	READY	STATUS	RESTARTS	AGE
temp	0/1	Pending	0	0s
temp	0/1	Pending	0	0s
temp	0/1	ContainerCreating	0	0s
temp	0/1	ErrImagePull	0	5s
temp	0/1	Terminating	0	6s
temp	0/1	Terminating	0	6s
temp	0/1	Terminating	0	7s
temp	0/1	Terminating	0	7s
temp	0/1	Terminating	0	7s
temp	0/1	Terminating	0	7s
temp	0/1	Terminating	0	7s

SS4:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG$ kubectl logs citizen -n pes2ug23cs044-alpha
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/10 15:53:58 [notice] 1#1: using the "epoll" event method
2026/02/10 15:53:58 [notice] 1#1: nginx/1.29.5
2026/02/10 15:53:58 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/02/10 15:53:58 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/02/10 15:53:58 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/02/10 15:53:58 [notice] 1#1: start worker processes
2026/02/10 15:53:58 [notice] 1#1: start worker process 29
2026/02/10 15:53:58 [notice] 1#1: start worker process 30
2026/02/10 15:53:58 [notice] 1#1: start worker process 31
2026/02/10 15:53:58 [notice] 1#1: start worker process 32
2026/02/10 15:53:58 [notice] 1#1: start worker process 33
2026/02/10 15:53:58 [notice] 1#1: start worker process 34
2026/02/10 15:53:58 [notice] 1#1: start worker process 35
2026/02/10 15:53:58 [notice] 1#1: start worker process 36
2026/02/10 15:53:58 [notice] 1#1: start worker process 37
2026/02/10 15:53:58 [notice] 1#1: start worker process 38
2026/02/10 15:53:58 [notice] 1#1: start worker process 39
2026/02/10 15:53:58 [notice] 1#1: start worker process 40
2026/02/10 15:53:58 [notice] 1#1: start worker process 41
2026/02/10 15:53:58 [notice] 1#1: start worker process 42
2026/02/10 15:53:58 [notice] 1#1: start worker process 43
2026/02/10 15:53:58 [notice] 1#1: start worker process 44

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG$
```

SS5:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl run broken --image=busybox --restart=Never --command -- sh -c "echo 'CRASH: something went wrong' >&2; exit 1"
pod/broken created

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl logs broken
'CRASH: something went wrong'

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$
```

SS6:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl create deployment pes2ug23cs044 --image=nginx
deployment.apps/pes2ug23cs044 created

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl get deployments -n pes2ug23cs044-alpha
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
pes2ug23cs044 1/1     1            1           28s

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
broken                             0/1     Error     0           3m20s
citizen                            1/1     Running   0           7m41s
pes2ug23cs044-7df87977bb-ghrwf     1/1     Running   0           42s
```

SS7:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl scale deployment pes2ug23cs044 --replicas=5
deployment.apps/pes2ug23cs044 scaled

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
broken                             0/1     Error     0           4m45s
citizen                            1/1     Running   0           9m6s
pes2ug23cs044-7df87977bb-2vddn     1/1     Running   0           18s
pes2ug23cs044-7df87977bb-5qhhr     1/1     Running   0           18s
pes2ug23cs044-7df87977bb-7jtth     1/1     Running   0           18s
pes2ug23cs044-7df87977bb-ghrwf     1/1     Running   0           2m7s
pes2ug23cs044-7df87977bb-t8prs     1/1     Running   0           18s

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl scale deployment pes2ug23cs044 --replicas=3
deployment.apps/pes2ug23cs044 scaled

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
broken                             0/1     Error     0           5m8s
citizen                            1/1     Running   0           9m29s
pes2ug23cs044-7df87977bb-2vddn     1/1     Running   0           41s
pes2ug23cs044-7df87977bb-5qhhr     1/1     Running   0           41s
pes2ug23cs044-7df87977bb-ghrwf     1/1     Running   0           2m30s
```

SS8:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl get rs,po
NAME
replicaset.apps/pes2ug23cs044-7df87977bb    DESIRED    CURRENT    READY    AGE
3                                           3          3          5m55s

NAME                                READY    STATUS    RESTARTS    AGE
pod/broken                          0/1      Error     0            8m33s
pod/citizen                         1/1      Running   0            12m
pod/pes2ug23cs044-7df87977bb-2vddn  1/1      Running   0            4m6s
pod/pes2ug23cs044-7df87977bb-5qhhr  1/1      Running   0            4m6s
pod/pes2ug23cs044-7df87977bb-ghrwf  1/1      Running   0            5m55s

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl set image deployment/pes2ug23cs044 nginx=nginx:1.25
deployment.apps/pes2ug23cs044 image updated

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl get rs,po
NAME
replicaset.apps/pes2ug23cs044-7df87977bb    DESIRED    CURRENT    READY    AGE
3                                           3          3          6m33s
replicaset.apps/pes2ug23cs044-86db96fb88    1          1          0          9s

NAME                                READY    STATUS    RESTARTS    AGE
pod/broken                          0/1      Error     0            9m11s
pod/citizen                         1/1      Running   0            13m
pod/pes2ug23cs044-7df87977bb-2vddn  1/1      Running   0            4m44s
pod/pes2ug23cs044-7df87977bb-5qhhr  1/1      Running   0            4m44s
pod/pes2ug23cs044-7df87977bb-ghrwf  1/1      Running   0            6m33s
pod/pes2ug23cs044-86db96fb88-c9hw5  0/1      ContainerCreating  0          9s
```

SS9:

```
(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl rollout history deployment pes2ug23cs044
deployment.apps/pes2ug23cs044
REVISION  CHANGE-CAUSE
2          <none>
3          <none>

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl rollout history deployment pes2ug23cs044 --revision=2
deployment.apps/pes2ug23cs044 with revision #2
Pod Template:
  Labels:      app=pes2ug23cs044
               pod-template-hash=86db96fb88
  Containers:
    nginx:
      Image:      nginx:1.25
      Port:       <none>
      Host Port:  <none>
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations: <none>

(akshay@AKSHAY)~/mnt/c/Users/AKSHAY_AG
$ kubectl rollout history deployment pes2ug23cs044 --revision=3
deployment.apps/pes2ug23cs044 with revision #3
Pod Template:
  Labels:      app=pes2ug23cs044
               pod-template-hash=7df87977bb
  Containers:
    nginx:
      Image:      nginx
      Port:       <none>
      Host Port:  <none>
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations: <none>
```


SS10:

```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl apply -f nginx-pes2ug23cs044.yaml -n pes2ug23cs044-beta
deployment.apps/nginx-pes2ug23cs044 unchanged

(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl get deployments -n pes2ug23cs044-beta
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-pes2ug23cs044	3/3	3	3	2m24s

SS11:

```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl apply -f pes2ug23cs044-service.yaml
service/pes2ug23cs044-service created

(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl get services -n pes2ug23cs044-beta
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
pes2ug23cs044-service	ClusterIP	10.101.177.36	<none>	80/TCP	87s

```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl describe service pes2ug23cs044-service -n pes2ug23cs044-beta
```

Name: pes2ug23cs044-service
Namespace: pes2ug23cs044-beta
Labels: <none>
Annotations: <none>
Selector: app=nginx
Type: ClusterIP
IP Family Policy: SingleStack
IP Families: IPv4
IP: 10.101.177.36
IPs: 10.101.177.36
Port: <unset> 80/TCP
TargetPort: 80/TCP
Endpoints: 10.244.0.46:80,10.244.0.45:80,10.244.0.44:80
Session Affinity: None
Internal Traffic Policy: Cluster
Events: <none>

```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl get endpoints pes2ug23cs044-service -n pes2ug23cs044-beta
```

NAME	ENDPOINTS	AGE
pes2ug23cs044-service	10.244.0.44:80,10.244.0.45:80,10.244.0.46:80	2m6s

SS12:

```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl get endpoints pes2ug23cs044-service -n pes2ug23cs044-beta
NAME                               ENDPOINTS                                     AGE
pes2ug23cs044-service             10.244.0.44:80,10.244.0.45:80,10.244.0.46:80 2m6s

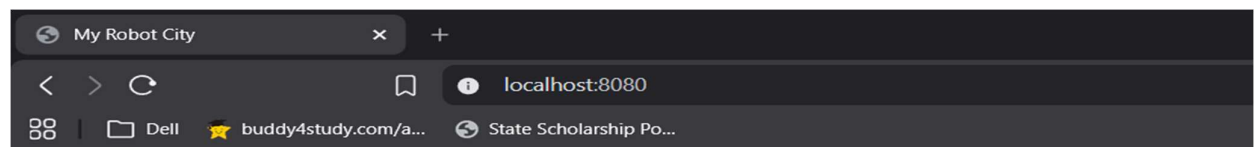
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl delete pod -l app=nginx -n pes2ug23cs044-beta
pod "nginx-pes2ug23cs044-5887598b5b-5rbz7" deleted from pes2ug23cs044-beta namespace
pod "nginx-pes2ug23cs044-5887598b5b-ctlzt" deleted from pes2ug23cs044-beta namespace
pod "nginx-pes2ug23cs044-5887598b5b-r4lpv" deleted from pes2ug23cs044-beta namespace

(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl get endpoints pes2ug23cs044-service -n pes2ug23cs044-beta
NAME                               ENDPOINTS                                     AGE
pes2ug23cs044-service             10.244.0.47:80,10.244.0.48:80,10.244.0.49:80 4m44s

(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl describe service pes2ug23cs044-service -n pes2ug23cs044-beta
Name:                             pes2ug23cs044-service
Namespace:                         pes2ug23cs044-beta
Labels:                            <none>
Annotations:                       <none>
Selector:                          app=nginx
Type:                              ClusterIP
IP Family Policy:                  SingleStack
IP Families:                       IPv4
IP:                                10.101.177.36
IPs:                               10.101.177.36
Port:                              <unset> 80/TCP
TargetPort:                        80/TCP
Endpoints:                         10.244.0.47:80,10.244.0.48:80,10.244.0.49:80
Session Affinity:                  None
Internal Traffic Policy:           Cluster
Events:                            <none>
```

SS13:

```
(akshay@AKSHAY)-[/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl port-forward deployment/nginx-pes2ug23cs044 8080:80 -n pes2ug23cs044-beta
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
```



Hello from district PES1UG23CSXXX-beta running in Kubernetes

SS14:

```
(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl delete namespace pes2ug23cs044-alpha
namespace "pes2ug23cs044-alpha" deleted

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl delete namespace pes2ug23cs044-beta
namespace "pes2ug23cs044-beta" deleted

(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG/PES2UG23CS044]
$ kubectl get ns
NAME                STATUS    AGE
default             Active   7h23m
kube-node-lease     Active   7h23m
kube-public         Active   7h23m
kube-system         Active   7h23m
```